ETAS RTA-BSW v12.7.0

RTA-Hwd (Eth) Reference Guide

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1 Ethernet Tranceiver

Module name	EthTrcv
Package	RTA-HWD (Eth)
AUTOSAR Module ID	73
AUTOSAR Version	23-11

1.1 Introduction

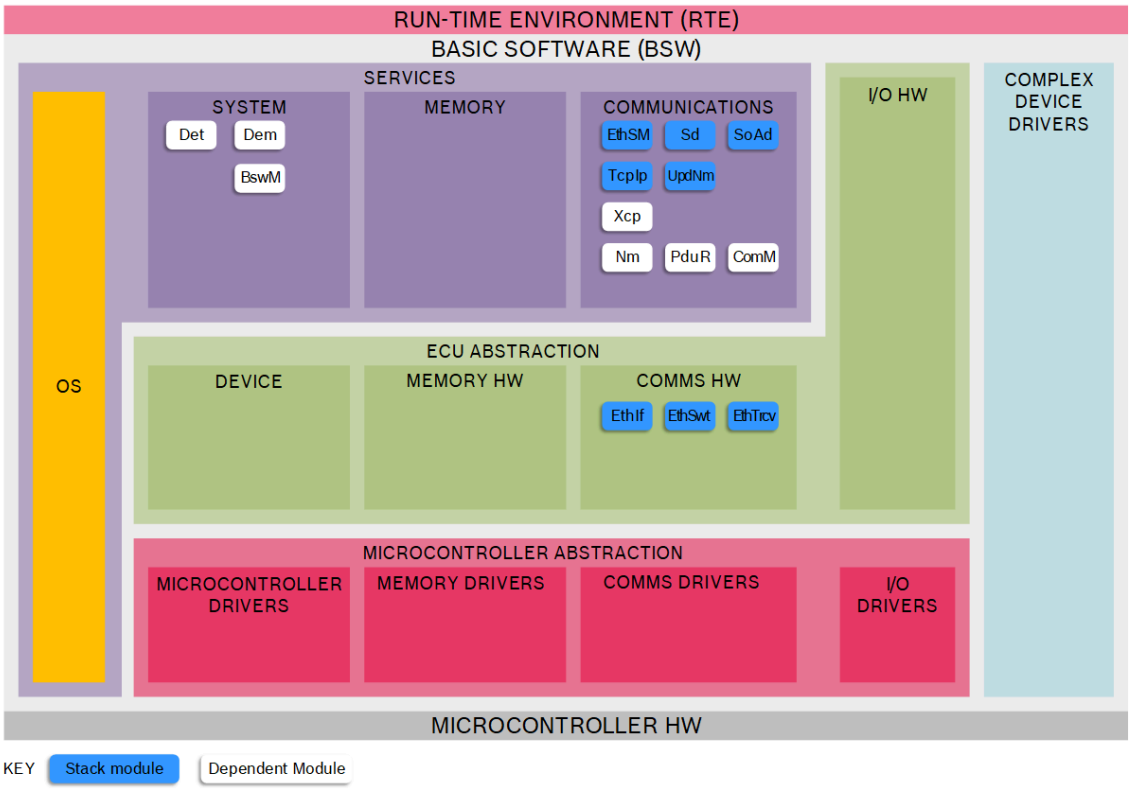

Figure 1.1.RTA-ETH Architecture

The Ethernet Transceiver Driver (EthTrcv) is located in the communication hardware abstraction layer, between the communication drivers layer and the upper communication services layer, and it forms part of the RTA-HWD (Eth) stack.

The diagram above shows the position of the EthTrcv module in the AUTOSAR layered architecture.

EthTrcv provides a hardware-independent interface comprising multiple physical Ethernet Transceivers to the upper layer Ethernet Interface (EthIf). This interface is uniform to all physical Ethernet Transceivers, allowing the upper layer to access the underlying bus system in a uniform manner.

The main tasks of the EthTrcv module include:

- Changing the state of the hardware transceiver

- Reading the current state of the hardware transceiver
- Detecting errors and the linkstate on the Ethernet line

The functionality provided by EthTrcv differs for each hardware transceiver - see the Supported Features section for more details.

1.1.1 Other Interaction/Dependencies

Eth

- The Eth driver configures the MDIO management communication path. It provides the Eth_ReadMii and Eth_WriteMii APIs for EthTrcv, allowing it to access and initialize the physical Ethernet transceiver hardware.

EthIf

- The EthIf module uses EthTrcv and calls its initialization APIs. In addition, EthTrcv calls the EthIf_TrcvModeIndication callback API to indicate Trcv mode changes to EthIf.

BswM

- The BSW Mode Manager (BswM) is used by the Ethernet Transceiver Driver for scheduling data consistency mechanisms and main function handling.

Port

- The Port driver configures the GPIO port pins (MDC and MDIO) used for the EthTrcv driver as input or output.

Module Configuration

This module can be generated by the RTA-BSW Configuration Generator.

For any unsupported automatic configuration, the module can be manually configured by the user.

1.2 Supported Features

- Getting/Setting the mode of the transceiver
- Checking the wakeup mode of the transceiver
- Monitoring the link status
- Auto Negotiation
- Getting the Baud Rate
- MDIO interactions (incl. Async)
- Running the diagnostic services

> **NOTE**
>
> Some functionality of EthTrcv is hardware-dependent and may vary according to the vendor-specific transceiver type being accessed.

1.2.1 Supported Hardware Devices

EthTrcv supports the following hardware devices:

Manufacturer	Device Name
NXP Semiconductors	TJA1100
Texas Instruments	DP83822
Marvell	88Q2112
Realtek	RTL9000AN
Realtek	RTL9000AA
Realtek	RTL9010AA

1.2.2 Supported Diagnostic Features

The sub-sections below describe the Diagnostic Features that are supported by some of the above transceiver types. See the limitations section for information on the diagnostic features that are not currently supported by the various devices.

1.2.2.1 For TJA1100

The following diagnostic features are supported for the TJA1100:

- 100Base-T1 Phy Test Modes
- Cable diagnostic
- Read and Write of Phy registers
- EthTrcv_GetBaudRate
- EthTrcv_GetDuplexMode
- EthTrcv_GetPhySignalQuality
- EthTrcv_SetPhyLoopbackMode
- EthTrcv_GetPhyIdentifier
- EthTrcv_SetPhyTxMode (ETHTRCV_PHYTXMODE_NORMAL and ETHTRCV_PHYTXMODE_TX_OFF)
- EthTrcv_GetLinkStatistics
- EthTrcv_GetLinkTrainingDuration
- EthTrcv_SetConnNegConfig (A non-Autosar API that is used to configure the PHY as Master or Slave).

- EthTrcv_GetConnNegConfig (A non-Autosar API that provides information on whether the PHY is configured as Master or Slave).

1.2.2.2 For RTL9000AA/RTL9000AN

The following diagnostic features are supported for the RTL9000AA and RTL9000AN:

- 100Base-T1 Phy Test and LoopbackModes
- EthTrcv_SetPhyLoopbackMode
- Cable diagnostic
- EthTrcv_GetPhyIdentifier
- EthTrcv_GetPhySignalQuality
- EthTrcv_SetPhyTxMode (ETHTRCV_PHYTXMODE_NORMAL)
- EthTrcv_SetConnNegConfig (A non-Autosar API that is used to configure the PHY as Master or Slave).
- EthTrcv_GetConnNegConfig (A non-Autosar API that provides information on whether the PHY is configured as Master or Slave).

1.3 Extensions

1.3.1 Configuration Extensions

The following configuration parameters are available in version v12.7.0 in addition to those defined in AR v22-11

1.3.1.1 EthTrcv/EthTrcvConfigSet/EthTrcvConfig/EthTrcvDemEventParameterRefs/EthTrcvRbDynCreate_ETHTRCV_E_ACCESS

Short Name	EthTrcvRbDynCreate_ETHTRCV_E_ACCESS
Description	Selects the way how the example DemEvent is created. If not set or set to Disabled, no such DemEvent is automatically created. If the dynamic creation of this DemEvent is enabled, the reference to this DemEvent (in parameter rba_Example_DemEventRef) is also automatically created and shall therefore not be set manually.
Multiplicity	0..1
Default Value	-
Enumeration Literals	Disabled, Standard
Type	ECUC-ENUMERATION-PARAM-DEF

1.3.1.2 EthTrcv/EthTrcvConfigSet/EthTrcvConfig/EthTrcvRbAccess

Short Name	EthTrcvRbAccess

Description	Container gives EthTrcv Driver information about access to a single Eth transceiver.
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

1.3.1.3 EthTrcv/EthTrcvConfigSet/EthTrcvConfig/EthTrcvRbAccess/EthTrcvRbDioAccess

Short Name	EthTrcvRbDioAccess
Description	Container gives Eth transceiver driver information about accessing ports and port pins. In addition relation between Eth transceiver hardware pin names and Dio port access information is given. If a Eth transceiver hardware has no Dio interface, there is no instance of this container.
Multiplicity	1..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

1.3.1.4 EthTrcv/EthTrcvConfigSet/EthTrcvConfig/EthTrcvRbAccess/EthTrcvRbDioAccess/EthTrcvRbDioChannelAccess

Short Name	EthTrcvRbDioChannelAccess
Description	Container gives DIO channel access by single Eth transceiver channel.
Multiplicity	1..3
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

1.3.1.5 EthTrcv/EthTrcvConfigSet/EthTrcvConfig/EthTrcvRbAccess/EthTrcvRbDioAccess/EthTrcvRbDioChannelAccess/EthTrcvRbDio-HardwareInterfaceType

Short Name	EthTrcvRbDioHardwareInterfaceType
Description	Container gives DIO channel access by single Eth transceiver channel.
Multiplicity	1..1
Default Value	-
Enumeration Literals	Reset, Enable, Wake
Type	ECUC-ENUMERATION-PARAM-DEF

1.3.1.6 EthTrcv/EthTrcvConfigSet/EthTrcvConfig/EthTrcvRbAccess/EthTrcvRbDioAccess/EthTrcvRbDioChannelAccess/EthTrcvRb-DioSymNameRef

Short Name	EthTrcvRbDioSymNameRef
Description	Container gives DIO channel access by single Eth transceiver channel.
Multiplicity	1..1
Default Value	-
Type	ECUC-REFERENCE-DEF

1.3.1.7 EthTrcv/EthTrcvConfigSet/EthTrcvConfig/EthTrcvRbDriverMiiApi

Short Name	EthTrcvRbDriverMiiApi
Multiplicity	1..1
Default Value	-
Enumeration Literals	ETHTRCV_DRIVER_SUPPORT_ASYNC_MIICOMMUNI-CATION, ETHTRCV_DRIVER_SUPPORT_NO_ASYNC_MIICOMMUNICATION, ETHTRCV_DRIVER_SUPPORT_OAWAPI_TO_PROVIDE_MII_FUNCTIONALITY, ETHTRCV_DRIVER_SUPPORT_PSEUDO_ASYNC_MIICOMMUNICATION
Type	ECUC-ENUMERATION-PARAM-DEF

Description: The communication from EthTrcv to Phy-hardware is managed by Mac (controller) hardware and SW. The SW-API is defined by Autosar by Eth_ReadMii and Eth_WriteMii. To avoidblocking of micro-controller during Mii-(MDio)-communication this API is defined to be asynchron. But some controllers does not support interrupts (by Mac- hardware)when a Mii(Mdio) word is written/received. These SW-controllers-drivers may not support asynchron communication. With this (EthTrcvDriverMiiAPI)configuration parameter it is possible to define the capability of Eth-Driver:ETHTRCV_DRIVER_SUPPORT-_OAWAPI_TO_PROVIDE_MII_FUNCTIONALITY is the default configuraion parameter. It is defined at RB that each Eth-controller-driver shall provides its capabilityto communicate async or sync by OAW-API: String Eth_Aspect_MiiBehaviour(). EthTrcv uses this OAW-API so configurator does not need to know the features of the EthDriver. If there is an error at validationin EthTrcv, Eth-driver is not RB conform and then configurator needs to know the capability of Eth-driver and should configure other parameters here (EthTrcvDriverMiiApi).ETHTRCV_-DRIVER_SUPPORT_ASYNC_MIICOMMUNICATION: Driver supports async communication and calls EthTrcv_ReadMiiIndication or EthTrcv WriteMiiIndication function after each Eth_ReadMii and Eth_WriteMii.ETHTRCV_DRIVER_SUPPORT-_NO_ASYNC_MIICOMMUNICATION: Driver does not support async communication and is not allowed to call EthTrcv_ReadMiiIndication nor EthTrcv WriteMiiIndication.

1.3.1.8 EthTrcv/EthTrcvConfigSet/EthTrcvConfig/EthTrcvRbHardware-Variant

Short Name	EthTrcvRbHardwareVariant
Description	RB (Vendor06) specific parameter:This parameter is applicable to choose a variant of Marvell Ash transceiver
Multiplicity	0..1
Default Value	-
Enumeration Literals	STDAUTOMPHY_MARVELL_88Q2220M_B0, STDAUTOMPHY_MARVELL_88Q2220M_B2, STDAUTOMPHY_MARVELL_88Q2221M_B0, STDAUTOMPHY_MARVELL_88Q2221M_B2
Type	ECUC-ENUMERATION-PARAM-DEF

1.3.1.9 EthTrcv/EthTrcvConfigSet/EthTrcvConfig/EthTrcvRbIoPowerSelection

Short Name	EthTrcvRbIoPowerSelection
Description	The RTL9000AA/RTL9000AN supports 3.3V/2.5V/1.8V I/O voltage of the MII interface.When using 3.3V as the power source,it must be provided be an external power source.The RTL9000AA/AN also supports an internal LDO for 1.8V/2.5V I/O voltage with some changes ofsettings.This 1.8V/2.5V power source could also be provided by an external power source.
Multiplicity	0..1
Default Value	-
Enumeration Literals	EXTERNAL_3_3V, INTERNAL_LDO_2_5V, EXTERNAL_2_5V, INTERNAL_LDO_1_8V, EXTERNAL_1_8V
Type	ECUC-ENUMERATION-PARAM-DEF

1.3.1.10 EthTrcv/EthTrcvConfigSet/EthTrcvConfig/EthTrcvRbPortMacLayerType

Short Name	EthTrcvRbPortMacLayerType
Description	Defines the MAC layer type of the ethernet transceiver.
Multiplicity	0..1
Default Value	-
Enumeration Literals	TRCV_MAC_LAYER_TYPE_MII, TRCV_MAC_LAYER_TYPE_RMII, TRCV_MAC_LAYER_TYPE_RevMII, TRCV_MAC_LAYER_TYPE_GMII, TRCV_MAC_LAYER_TYPE_RGMII, TRCV_MAC_LAYER_TYPE_RGMII_DELAY, TRCV_MAC_LAYER_TYPE_SGMII

Type	ECUC-ENUMERATION-PARAM-DEF

1.3.1.11 EthTrcv/EthTrcvConfigSet/EthTrcvConfig/EthTrcvRbSplitInitInto-Blocks

Short Name	EthTrcvRbSplitInitIntoBlocks
Description	When this parameter is true the Init commands are split into blocks.The first block isexecuted from trcv init api and remaining blocks are executed from main functionand hence takes more time.When this parameter is false the init commands are executed frominit api.The default value of parameter is false.
Multiplicity	0..1
Default Value	-
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.12 EthTrcv/EthTrcvConfigSet/EthTrcvConfig/EthTrcvRbTrcvSpecific-Param

Short Name	EthTrcvRbTrcvSpecificParam
Description	Transceiver specific parameters.
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

1.3.1.13 EthTrcv/EthTrcvConfigSet/EthTrcvConfig/EthTrcvRbTrcvSpecific-Param/EthTrcvRbTJA1100

Short Name	EthTrcvRbTJA1100
Description	Special Container for TJA1100 configuration. This can be configured for enabling some of the features of the TJA1100. But this has to be done in accordance with the hardware configuration.
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

1.3.1.14 EthTrcv/EthTrcvConfigSet/EthTrcvConfig/EthTrcvRbTrcvSpecific-Param/EthTrcvRbTJA1100/EthTrcvRbAUTO_PWD

Short Name	EthTrcvRbAUTO_PWD

Description	RB (Vendor06) specific parameter: If autonomous power-down is enabled via Configuration register 1 (AUTO_PWD = 1), the TJA1100 goes to Sleep Request mode automatically if no Ethernet frames have been received at the MDI and MII for the time-out time, tto(pd)autn. Currently, it is not supported
Multiplicity	0..1
Default Value	ETHTRCV_AUTO_PWD_DISABLED
Enumeration Literals	ETHTRCV_AUTO_PWD_DISABLED, ETHTRCV_AUTO_PWD_ENABLED
Type	ECUC-ENUMERATION-PARAM-DEF

1.3.1.15 EthTrcv/EthTrcvConfigSet/EthTrcvConfig/EthTrcvRbTrcvSpecificParam/EthTrcvRbTJA1100/EthTrcvRbCONFIG_INH

Short Name	EthTrcvRbCONFIG_INH
Description	RB (Vendor06) specific parameter: Reg17 Bit1, 0 = INH switched off in Disable mode; 1 = INH switched on in Disable mode.This control signal is used to activate the supply to the ECU's microcontroller.
Multiplicity	0..1
Default Value	ETHTRCV_CONFIG_INH_ON_IN_DIS_MODE
Enumeration Literals	ETHTRCV_CONFIG_INH_OFF_IN_DIS_MODE, ETHTRCV_CONFIG_INH_ON_IN_DIS_MODE
Type	ECUC-ENUMERATION-PARAM-DEF

1.3.1.16 EthTrcv/EthTrcvConfigSet/EthTrcvConfig/EthTrcvRbTrcvSpecificParam/EthTrcvRbTJA1100/EthTrcvRbJUMBO_ENABLE

Short Name	EthTrcvRbJUMBO_ENABLE
Description	RB (Vendor06) specific parameter: It enables/disables the jumbo packet support. Reg 19 Bit2: 0 = packets up to 4 kB supported; 1 = packets up to 16 kB supported
Multiplicity	0..1
Default Value	ETHTRCV_16KB_SUPPORT
Enumeration Literals	ETHTRCV_4KB_SUPPORT, ETHTRCV_16KB_SUPPORT
Type	ECUC-ENUMERATION-PARAM-DEF

1.3.1.17 EthTrcv/EthTrcvConfigSet/EthTrcvConfig/EthTrcvRbTrcvSpecificParam/EthTrcvRbTJA1100/EthTrcvRbLED_ENABLE

Short Name	EthTrcvRbLED_ENABLE

Description	RB (Vendor06) specific parameter: It enables/disables the LED output and Wake input. This shall be configured according to the hardware configuration. Reg 18 Bit3: 0 = LED output disabled, WAKE input enabled; 1 = LED output enabled, WAKE input disabled.
Multiplicity	0..1
Default Value	ETHTRCV_LED_OUTPUT_DISABLED
Enumeration Literals	ETHTRCV_LED_OUTPUT_DISABLED, ETHTRCV_LED_OUTPUT_ENABLED
Type	ECUC-ENUMERATION-PARAM-DEF

1.3.1.18 EthTrcv/EthTrcvConfigSet/EthTrcvConfig/EthTrcvRbTrcvSpecificParam/EthTrcvRbTJA1100/EthTrcvRbLED_MODE

Short Name	EthTrcvRbLED_MODE
Description	RB (Vendor06) specific parameter: This configuration is valid only if the EthTrcvRbLED_ENABLE configuration is enabled.The LED can be configured to indicate link status,frame reception,symbol error or CRS signal.
Multiplicity	0..1
Default Value	ETHTRCV_LED_MODE_LINK_UP
Enumeration Literals	ETHTRCV_LED_MODE_LINK_UP, ETHTRCV_LED_MODE_FRAME_RECEPTION, ETHTRCV_LED_MODE_SYMBOL_ERROR, ETHTRCV_LED_MODE_CRS_SIGNAL
Type	ECUC-ENUMERATION-PARAM-DEF

1.3.1.19 EthTrcv/EthTrcvConfigSet/EthTrcvConfig/EthTrcvRbTrcvSpecificParam/EthTrcvRbTJA1100/EthTrcvRbLinkUpDelay

Short Name	EthTrcvRbLinkUpDelay
Description	Specifies the delay in microseconds between mode active and enabling link control.
Multiplicity	0..1
Value Range	2000..5000
Default Value	2000
Type	ECUC-INTEGER-PARAM-DEF

1.3.1.20 EthTrcv/EthTrcvConfigSet/EthTrcvConfig/EthTrcvRbTrcvSpecificParam/EthTrcvRbTJA1100/EthTrcvRbMII_DRIVER

Short Name	EthTrcvRbMII_DRIVER
Description	RB (Vendor06) specific parameter: This configures the driver strength for the MII lines.

Multiplicity	0..1
Default Value	ETHTRCV_MII_DRIVER_STANDARD
Enumeration Literals	ETHTRCV_MII_DRIVER_STANDARD, ETHTRCV_MII_DRIVER_REDUCED
Type	ECUC-ENUMERATION-PARAM-DEF

1.3.1.21 EthTrcv/EthTrcvConfigSet/EthTrcvConfig/EthTrcvRbTrcvSpecificParam/EthTrcvRbTJA1100/EthTrcvRbOperationMode

Short Name	EthTrcvRbOperationMode
Description	RB (Vendor06) specific parameter:In managed operation,once the mode of the transceiver is active, the LINK_CONTROL bit needs to be set to 'ENABLE',the PHY configured as master initiates the training sequence by transmitting idle pulses.The link is established when bit LINK_UP in the communication status register is set.In Autonomous operation, the link is activated by the transceiver once the transceivers mode is active.
Multiplicity	1..1
Default Value	-
Enumeration Literals	ETHTRCV_AUTO_OP_MANAGED, ETHTRCV_AUTO_OP_AUTONOMOUS
Type	ECUC-ENUMERATION-PARAM-DEF

1.3.1.22 EthTrcv/EthTrcvConfigSet/EthTrcvConfig/EthTrcvRbTrcvSpecificParam/EthTrcvRbTJA1100/EthTrcvRbTX_AMPLITUDE

Short Name	EthTrcvRbTX_AMPLITUDE
Description	RB (Vendor06) specific parameter:The default value of 1000 mV can support a link of up to 15 m, while the lower values of 500 mV and 750 mV may be sufficient for shorter links of up to 2m.For links longer than 15 m and those exposed to strong electromagnetic interference, 1250 mV is recommended to cope with the interference.
Multiplicity	0..1
Default Value	ETHTRCV_TX_AMPLITUDE_1000MV
Enumeration Literals	ETHTRCV_TX_AMPLITUDE_500MV, ETHTRCV_TX_AMPLITUDE_750MV, ETHTRCV_TX_AMPLITUDE_1000MV, ETHTRCV_TX_AMPLITUDE_1250MV
Type	ECUC-ENUMERATION-PARAM-DEF

1.3.1.23 EthTrcv/EthTrcvConfigSet/EthTrcvConfig/EthTrcvRbWakeUpDuration

Short Name	EthTrcvRbWakeUpDuration
Description	Specifies the Wakeup High or Low Duration time configurable at HW. This parameter is supported only for RTL9010AA.
Multiplicity	0..1
Default Value	-
Enumeration Literals	TRCV_WUD_20_US, TRCV_WUD_10_MS
Type	ECUC-ENUMERATION-PARAM-DEF

1.3.1.24 EthTrcv/EthTrcvConfigSet/EthTrcvConfig/EthTrcvRbWakeUpEdgeDetection

Short Name	EthTrcvRbWakeUpEdgeDetection
Description	Specifies the Wakeup High or Low edge detection configurable at HW. This parameter is supported only for RTL9010AA.
Multiplicity	0..0
Default Value	-
Enumeration Literals	TRCV_WUD_EDGE_HIGH, TRCV_WUD_EDGE_LOW
Type	ECUC-ENUMERATION-PARAM-DEF

1.3.1.25 EthTrcv/EthTrcvConfigSet/EthTrcvConfig/EthTrcvWakeupMap/EthTrcvRbWakeupReason

Short Name	EthTrcvRbWakeupReason
Description	This parameter defines the transceiver wake up reasons.
Multiplicity	1..1
Default Value	-
Enumeration Literals	ETHTRCV_WUR_BUS, ETHTRCV_WUR_GENERAL, ETHTRCV_WUR_INTERNAL, ETHTRCV_WUR_PIN, ETHTRCV_WUR_POWER_ON, ETHTRCV_WUR_RESET, ETHTRCV_WUR_SYSERR
Type	ECUC-ENUMERATION-PARAM-DEF

1.3.1.26 EthTrcv/EthTrcvConfigSet/EthTrcvConfig/EthTrcvWakeupMap/EthTrcvRbWakeupSourceRef

Short Name	EthTrcvRbWakeupSourceRef
Description	Configures the wake-up source defined in EcuM.

Multiplicity	1..1
Default Value	-
Type	ECUC-REFERENCE-DEF

1.3.1.27 EthTrcv/EthTrcvGeneral/EthTrcvGetPhySignalQualityApi

Short Name	EthTrcvGetPhySignalQualityApi
Description	Enables / Disables EthTrcv_GetPhySignalQuality API.
Multiplicity	1..1
Default Value	-
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.28 EthTrcv/EthTrcvGeneral/EthTrcvRbCallBackHeaderFile

Short Name	EthTrcvRbCallBackHeaderFile
Description	Specifies the header file, where the call back API(Configured via parameter - EthTrcvRbCableDiagCallout) is declared. Needs to be configured if the Transceiver selected is RTL9000AA
Multiplicity	0..1
Default Value	-
Type	ECUC-STRING-PARAM-DEF

1.3.1.29 EthTrcv/EthTrcvGeneral/EthTrcvRbDriverPrefix

Short Name	EthTrcvRbDriverPrefix
Description	RB (Vendor06) specific parameter: Selection of EthernetTransceiver-Driver to support Multiple Transceiver.
Multiplicity	0..1
Default Value	-
Enumeration Literals	EthTrcv_11_MacPhy, EthTrcv_11_MultiGigPhy, EthTrcv_11_StdAutomPhy
Type	ECUC-ENUMERATION-PARAM-DEF

1.3.1.30 EthTrcv/EthTrcvGeneral/EthTrcvRbDriverPrefixHeaderFile

Short Name	EthTrcvRbDriverPrefixHeaderFile
Description	Specifies the header file, where Eth_ReadMii and Eth_WriteMii APIs are available when there is no ethIfEthDriverPrefix available in the EthIf. Mandatory configuration in this case.
Multiplicity	0..1

Default Value	-
Type	ECUC-STRING-PARAM-DEF

1.3.1.31 EthTrcv/EthTrcvGeneral/EthTrcvRbEnableGMNodeIdReMpg

Short Name	EthTrcvRbEnableGMNodeIdReMpg
Description	This RB parameter enables the GM specific NodeID remapping feature.
Introduction	* true: NodeID remapping feature is enabled. * false: NodeID remapping feature is disabled.
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.32 EthTrcv/EthTrcvGeneral/EthTrcvRbGetConnNegConfigApi

Short Name	EthTrcvRbGetConnNegConfigApi
Description	Enables / Disables EthTrcvRbGetConnNegConfig API.
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.33 EthTrcv/EthTrcvGeneral/EthTrcvRbGetLinkStatisticsApi

Short Name	EthTrcvRbGetLinkStatisticsApi
Description	Enables / Disables EthTrcvRbGetLinkStatistics API.
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.34 EthTrcv/EthTrcvGeneral/EthTrcvRbGetLinkTrainingDurationApi

Short Name	EthTrcvRbGetLinkTrainingDurationApi
Description	Enables / Disables EthTrcvRbGetLinkTrainingDuration API.
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.35 EthTrcv/EthTrcvGeneral/EthTrcvRbHardwareType

Short Name	EthTrcvRbHardwareType

Description	RB (Vendor06) specific parameter: Type of the used Hardware
Multiplicity	1..1
Default Value	-
Enumeration Literals	DP83822, RTL9010AA, TJA1100, RTL9000AA, RTL9000AN, MARVELL_88Q2112, AD330x, NCV7410, MARVELL_88Q324x, MARVELL_88Q222x
Type	ECUC-ENUMERATION-PARAM-DEF

1.3.1.36 EthTrcv/EthTrcvGeneral/EthTrcvRbLinkTrainingCallout

Short Name	EthTrcvRbLinkTrainingCallout
Description	Specifies the callback that needs to be called on completion of Link training. Only requireed for TJA1100 Syntax: void <CallBackName> (uint8 TrcvIdx, uint32 LinkTrainingDuration TrcvIdx - Id of the Transceiver LinkTrainingDuration - Link training duration in milli seconds
Multiplicity	0..1
Default Value	-
Type	ECUC-STRING-PARAM-DEF

1.3.1.37 EthTrcv/EthTrcvGeneral/EthTrcvRbPolarityCorrectionDisable

Short Name	EthTrcvRbPolarityCorrectionDisable
Description	This enables or disables polarity correction. Polarity correction supported only for RTL9000AA and RTL9000AN.
Multiplicity	0..1
Default Value	FALSE
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.38 EthTrcv/EthTrcvGeneral/EthTrcvRbRwakeDis

Short Name	EthTrcvRbRwakeDis
Description	Enables or disables remote wake up pulse to link partner. Configure this parameter to true if Link partner does not support TC10 to disable default remote wake pulse upon local wake up to RTL9000AA/AN. This parameter is supported only for RTL9000AA/AN.
Multiplicity	0..1
Default Value	FALSE
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.39 EthTrcv/EthTrcvGeneral/EthTrcvRbSetConnNegConfigApi

Short Name	EthTrcvRbSetConnNegConfigApi
Description	Enables / Disables EthTrcvRbSetConnNegConfig API.
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.40 EthTrcv/EthTrcvGeneral/EthTrcvRbSetPhyLoopbackModeApi

Short Name	EthTrcvRbSetPhyLoopbackModeApi
Description	This api configures the Phy to requested loopback(internal, external or remote) mode. This api is available for TJA1100, RTL9000AA and RTL9010AA EthTrcv. Enables / Disables EthTrcvRbSetPhyLoopbackMode API.
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.41 EthTrcv/EthTrcvGeneral/EthTrcvSetPhyTxModeApi

Short Name	EthTrcvSetPhyTxModeApi
Description	Enables / Disables EthTrcv_SetPhyTxMode API.
Multiplicity	1..1
Default Value	-
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.2 Additional Services

The following API are available in version v12.7.0 in addition to those defined in AR v22-11

1.3.2.1 EthTrcv_TransceiverInit

Syntax	`EthTrcv_TransceiverInit(uint8 TrcvIdx, uint8 CfgIdx)`
Parameters	`TrcvIdx` Identifies Transceiver to which the API has to be applied. `CfgIdx` Not used in current implementation.
Description	Configures the transceiver.

1.3.2.2 EthTrcv_SetConnNegConfig

Syntax	`EthTrcv_SetConnNegConfig(uint8 TrcvIdx, EthTrcv_ConnNegType_ten ConnNeg)`
Parameters	`TrcvIdx` Index of the transceiver on the MII. `ConnNeg` List of Values which can be provided by EthTrcv TJA1100,RTL9000AA and RTL9000AN.
Description	Used to configure the phy as Master or Slave. The MII behaviour shall be Sync.

1.3.2.3 EthTrcv_GetConnNegConfig

Syntax	`EthTrcv_GetConnNegConfig(uint8 TrcvIdx, EthTrcv_ConnNegType_ten* ConnNegPtr)`
Parameter In	`TrcvIdx` Index of the transceiver on the MII.
Parameter Out	`ConnNegPtr` List of Values which can be provided by EthTrcv TJA1100.
Description	Specifies whether the Phy is configured as Master or Slave. Synchronous, Non Reentrant.
Note	The MII behaviour shall be Sync.

1.3.2.4 EthTrcv_ReadPhyRegister

Syntax	`EthTrcv_ReadPhyRegister( VAR(uint8, AUTOMATIC) CtrlIdx_u8, VAR(uint8, AUTOMATIC) TrcvIdx_u8, VAR(uint8, AUTOMATIC) RegIdx_u8, P2VAR(uint16, AUTOMATIC, AUTOMATIC) RegValPtr_pu16 )`
Parameter In	`CtrlIdx_u8` Index of the controller within the context of the Ethernet Transceiver Driver.
Parameter In	`TrcvIdx_u8` Index of the transceiver on the MII.
Parameter In	`RegIdx_u8` Index of the transceiver register on the MII.
Parameter Out	`RegValPtr_pu16` Filled with the register content of the indexed register.
Description	Reads the indexed transceiver register value; uses Eth_ReadMii api provided by Eth. Service ID 0x53, Synchronous, Non Reentrant.
Note	The MII behaviour shall be Sync.

1.3.2.5 EthTrcv_WritePhyRegister

Syntax	`EthTrcv_WritePhyRegister( VAR(uint8, AUTOMATIC) CtrlIdx_u8, VAR(uint8, AUTOMATIC) TrcvIdx_u8, VAR(uint8, AUTOMATIC) RegIdx_u8, VAR(uint16, AUTOMATIC) RegVal_u16 )`
Parameter In	`CtrlIdx_u8` Index of the controller within the context of the Ethernet Driver.

Parameter In	`TrcvIdx_u8` Index of the transceiver on the MII.
Parameter In	`RegIdx_u8` Index of the transceiver register on the MII.
Parameter In	`RegVal_u16` Value to be written into the indexed register.
Description	Writes the indexed transceiver register with the input value; uses Eth_WriteMii api provided by Eth. Service ID 0x54, Synchronous, Non Reentrant.
Note	The MII behaviour of the Eth driver must be Sync. Once these apis are used to update registers of the transceiver, the Phy state cannot be determined by the EthTrcv driver and hence a re-initialization would be requried.

1.3.2.6 EthTrcv_GetLinkStatistics

Syntax	`EthTrcv_GetLinkStatistics( VAR (uint8, AUTOMATIC) TrcvIdx_u8, P2VAR(EthTrcv_LinkStatisticsType, AUTOMATIC, AUTOMATIC) LinkStats_pst)`
Parameter In	`TrcvIdx_u8` Index of the transceiver on the MII.
Parameter Out	`LinkStats_pst` Filled with the Link statistics values for the indexed register.
Description	Extracts the link statistics from the indexed transceiver. Service ID 0x55, Synchronous, Non Reentrant.
Note	The MII behaviour shall be Sync.

1.3.2.7 EthTrcv_GetLinkTrainingDuration

Syntax	`EthTrcv_GetLinkTrainingDuration( VAR (uint8, AUTOMATIC) TrcvIdx_u8)`
Parameter In	`TrcvIdx_u8` Index of the transceiver on the MII.
Description	Measures the link training duration. Only the link training is initiated by the api. The link training duration is calculated asynchronously and a callback is given when results are available. Service ID 0x56, Asynchronous, Non Reentrant.
Note	The MII behaviour shall be Sync.

1.4 Limitations

This section is divided into the following 3 sub-sections:

- General Limitations
- EthTrcv Feature-Specific Limitations
- Hardware-Specific Limitations

1.4.1 General Limitations

- DSM Re-init is not supported.
- Only Marvell88Q2112 and RTL9010AA support 1000Mbps with SGMII interface.
- The RTA-BSW implementation of the EthTrcv module does not support MCALs using Asynchronous Type (ASYNC).
- Only MII mode is tested for TJA1100, RTL9000AA and DP83822 Eth Phys.
- All the EthTrcv are tested only in MII mode except (1) Marvell88Q2112, which supports only RGMII and SGMII and (2) RTL9010AA, which supports only SGMII.
- No regression testing for any of the diagnostic features for TJA1100 and RTL9000AA.
- Currently only 100Mbps configuration is supported for all Phy hardware types.
- Only pre-compile time is supported for EthTrcv.

1.4.2 Feature-Specific Limitations

Phy Test Mode / Loopback Mode

- When one of the Phy test Modes or loopback Modes is enabled, then no other Diagnostic feature shall be requested until the test Mode or loopback Mode is disabled. (The TJA1100 and RTL9000AA hardware cannot run multiple Diagnostic features at a time).
- When any of the test Modes is enabled, the physical link goes to DOWN but the software link status is not made DOWN and instead remains at its previous status. This is required so that the TCP Socket connection to the Diagnostic tool is maintained. The physical link is activated again once the PhyTestModes are disabled.

EthTrcv_SetPhyLoopbackMode

- The External loopback mode requires the Link partner to be disconnected
- In Internal loopback mode, the frames transmitted from MAC will be re-transmitted to MAC but not forwarded to the Link partner.

EthTrcv_GetPhyIdentifier

- The 24 bit OUI number is stored in a uint32 variable, with Bit 23 of the OUI as Bit 0 of the uint32 variable, and Bit 0 of the OUI as Bit 23 of the uint32 variable. (Note: This is not clearly specified in Autosar).

1.4.3 Hardware-Specific Limitations

1.4.3.1 DP83822

The following features are not currently supported for the DP83822:

- The RMII mode of connection between the Eth MAC and the Eth Phy
- WakeUp mode functionalities
- Diagnostic features
- PTP functionalities
- Error handling. It is assumed that the EthTrcv hardware is enabled and that access to Phy registers is available, otherwise EthTrcv initialization will not be successful.

1.4.3.2 RTL9000AA/RTL9000AN

Note the following considerations with regard to the current support for the RTL9000AA and RTL9000AN:

Cable diagnostic

- The RTL9000AA and RTL9000AN support Cable diagnostic as per AR4.4
- The cable test requires 200ms to 300ms of time.
- When EthTrcv_GetCableDiagnosticResult is called, it only accepts the requests, and E_OK is returned with ResultPtr as ETHTRCV_CABLEDIAG_OK.

EthTrcv_GetPhySignalQuality

- The SQI measurement is done only when the cable is connected (between DUT and the link partner) and when the link is Active. The SQI measurement is not reliable when the link is DOWN.

100Base-T1 Phy Test Modes and Loopback Modes

When any of the test modes or loopback modes are enabled, the physical link goes to DOWN but the software link status is not made DOWN. It is maintained to the previous value only. This is required so that the TCP Socket connection to diagnostic tool is maintained. The physical link is activated again once the enabled diagnostic feature is disabled.

RTL9000AA/AN Operating Mode

The operating mode of RTL9000AA/AN at the end of initialization depends on ST_CHG pin value. The mode is normal for High and standby for Low. When the pin value is high multiple links up and down can be observed on Canoe (link partner) during initialization.

1.4.3.3 TJA1100

Cable diagnostic

- The TJA1100 MRA4 sample supports Cable diagnostic as per AR4.4 requirement.
- The TJA1100 MRA4 sample can detect only Open circuit and Short circuit in-and-between the cable wires. A short between a single wire, and the battery voltage or ground, may not be detected.

Read and Write of Phy registers

- Once EthTrcv_WritePhyRegister has been used to update any of the TJA1100 registers, the status of the Eth Phy cannot be determined by EthTrcv and a re-initialization will be required.

EthTrcv_GetLinkTrainingDuration

- The link training duration is available in milliseconds. Since there is no interrupt support for the link training feature, polling mode is used and the precision of this time would be the raster in which the main function is scheduled.

Phy Test Mode

When one of the Phy Test Modes are enabled, then no other diagnostic feature shall be requested until the test mode is disabled. The TJA1100 hardware cannot run mutiple diagnostic features at a time.

Phy Loopback Mode

When one of the Phy Loopback Modes are enabled, then no other diagnostic feature shall be requested untill the loopback mode is disabled. The TJA1100 hardware cannot run mutiple Diagnostic features at a time.

100Base-T1 Phy Test Modes

Whilst 100Base-T1 Phy Test Modes are supported. When any of the test modes are enabled, the physical link goes to DOWN but the software link status is not made DOWN. It is maintained to the previous value only. This is required so that the TCP Socket connection to diagnostic tool is maintained. The physical link is activated again once the PhyTestModes are disabled.

Link Start Up

The TJA1100 MRA3 sample has Link drop issues during link start up and also during normal operation, hence only MRA4 sample shall be used in the hardware. This is the recommendation from hardware vendor.

1.4.3.4 RTL9010AA V2

The following features are not currently supported for the RTL9010AA V2:

- EthTrcv_GetBaudRate
- EthTrcv_GetDuplexMode

- Cable test diagnosis yet to be tested completely for faults.
- WAKE UP support is disabled as thorough testing has yet to be completed.

- Auto Negotiation is not tested as CANOe does not support the Auto Negotiation option for T1.
- EthTrcv_GetConnNegConfig and EthTrcv_SetConnNegConfig APIs are not supported if Transceiver is in Auto Negotiation mode.

1.5 Exclusions

The following features from AR v22-11 are not supported in version v12.7.0.

API functions not currently implemented:

- `EthTrcv_GetBusWuReason`
- `EthTrcv_GetMacMethod`
- `EthTrcv_MacSecUpdateSecY`
- `EthTrcv_MacSecInitRxSc`
- `EthTrcv_MacSecResetRxSc`
- `EthTrcv_MacSecAddTxSa`
- `EthTrcv_MacSecUpdateTxSa`
- `EthTrcv_MacSecDeleteTxSa`
- `EthTrcv_MacSecAddRxSa`
- `EthTrcv_MacSecUpdateRxSa`
- `EthTrcv_MacSecDeleteRxSa`
- `EthTrcv_MacSecGetTxSaNextPn`
- `EthTrcv_MacSecGetMacSecStats`
- `EthTrcv_MacSecSetControlledPortEnabled`
- `EthTrcv_EthSwitchMacSecUpdateSecY`
- `EthTrcv_EthSwitchMacSecAddTxSa`
- `EthTrcv_SwitchMacSecAddRxSa`
- `EthTrcv_SwitchMacSecGetMacSecStats`
- `EthTrcv_SetTransceiverWakeupMode`
- `EthTrcv_GetTransceiverWakeupMode`

The following configuration parameters from AR v22-11 are not supported in version v12.7.0.

- EthTrcv/EthTrcvConfigSet/EthTrcvConfig/EthTrcvConfigEcucPartitionRef
- EthTrcv/EthTrcvConfigSet/EthTrcvConfig/EthTrcvForceSleepEnabled
- EthTrcv/EthTrcvConfigSet/EthTrcvConfig/EthTrcvIcuChannelRef
- EthTrcv/EthTrcvConfigSet/EthTrcvConfig/EthTrcvMacLayerSpeed
- EthTrcv/EthTrcvConfigSet/EthTrcvConfig/EthTrcvMacLayerSubType
- EthTrcv/EthTrcvConfigSet/EthTrcvConfig/EthTrcvMacLayerType

- EthTrcv/EthTrcvConfigSet/EthTrcvConfig/EthTrcvPhysLayerPlcaLocalNodeId
- EthTrcv/EthTrcvConfigSet/EthTrcvConfig/EthTrcvPhysLayerPlcaMaxBurstCount
- EthTrcv/EthTrcvConfigSet/EthTrcvConfig/EthTrcvPhysLayerPlcaMaxBurstTimer
- EthTrcv/EthTrcvConfigSet/EthTrcvConfig/EthTrcvPhysLayerPlcaNodeCount
- EthTrcv/EthTrcvConfigSet/EthTrcvConfig/EthTrcvPhysLayerPlcaTransmitOpportunityTimer
- EthTrcv/EthTrcvConfigSet/EthTrcvConfig/EthTrcvPhysLayerType
- EthTrcv/EthTrcvConfigSet/EthTrcvConfig/EthTrcvSleepRequestNumberOfRepetitions
- EthTrcv/EthTrcvConfigSet/EthTrcvConfig/EthTrcvSleepRequestRepetitionPeriod
- EthTrcv/EthTrcvConfigSet/EthTrcvConfig/EthTrcvWakeUpCallout
- EthTrcv/EthTrcvConfigSet/EthTrcvConfig/EthTrcvWakeupForwardLocalEnabled
- EthTrcv/EthTrcvConfigSet/EthTrcvConfig/EthTrcvWakeupForwardRemoteEnabled
- EthTrcv/EthTrcvConfigSet/EthTrcvConfig/EthTrcvWakeupLocalDetectionTime
- EthTrcv/EthTrcvConfigSet/EthTrcvConfig/EthTrcvWakeupLocalDurationTime
- EthTrcv/EthTrcvConfigSet/EthTrcvConfig/EthTrcvWakeupLocalEnabled
- EthTrcv/EthTrcvConfigSet/EthTrcvConfig/EthTrcvWakeupMap/EthTrcvWakeupReason
- EthTrcv/EthTrcvConfigSet/EthTrcvConfig/EthTrcvWakeupMap/EthTrcvWakeupSourceRef
- EthTrcv/EthTrcvConfigSet/EthTrcvConfig/EthTrcvWakeupRemoteEnabled
- EthTrcv/EthTrcvGeneral/EthTrcvEcucPartitionRef
- EthTrcv/EthTrcvGeneral/EthTrcvEnablePLCA
- EthTrcv/EthTrcvGeneral/EthTrcvGetTransceiverWakeupModeApi

1.6 Deviations

1.6.1 Configuration Deviations

Configuration parameters which differ from the AR v22-11 specification are listed below.

1.6.1.1 EthTrcv/EthTrcvConfigSet/EthTrcvConfig/EthTrcvActAsSlavePassiveEnabled

[RTA-BSW] Description	Specifies if the ECU is acting as a passive communication slave on the corresponding ComM channel (corresponding ComM channel has ComMNMVariant set to SLAVE_PASSIVE). If the parameter is set to TRUE, the Ethernet transceiver driver shall poll the maintained Ethernet hardware for a signaled Sleep.Indication (according to OA TC10) in the context of the EthTrcv_MainFunction.
[AUTOSAR] Description	Specifies if the ECU is acting as a passive communication slave on the corresponding ComM channel (corresponding ComM channel has ComMNMVariant set to SLAVE_PASSIVE). Tags: atp.Status=draft
[REMOVED] Introduction	If the parameter is set to TRUE, the Ethernet transceiver driver shall poll the maintained Ethernet hardware for a signaled Sleep.Indication (according to OA TC10) in the context of the EthTrcv_MainFunction.

1.6.1.2 EthTrcv/EthTrcvConfigSet/EthTrcvConfig/EthTrcvConnNeg

[RTA-BSW] Enumeration Literals	TRCV_CONN_NEG_AUTO, TRCV_CONN_NEG_MASTER, TRCV_CONN_NEG_SLAVE, TRCV_CONN_NEG_NONE
[AUTOSAR] Enumeration Literals	TRCV_CONN_NEG_AUTO, TRCV_CONN_NEG_MASTER, TRCV_CONN_NEG_NONE, TRCV_CONN_NEG_SLAVE

1.6.1.3 EthTrcv/EthTrcvConfigSet/EthTrcvConfig/EthTrcvMgmtInterface/EthTrcvMiiInterface

[REMOVED] Introduction	Note: In case of external MACPHY use this parameter may be enabled.

1.6.1.4 EthTrcv/EthTrcvConfigSet/EthTrcvConfig/EthTrcvMgmtInterface/EthTrcvSwitchInterface/EthTrcvSwitchPortRef

[RTA-BSW] Lower Multiplicity	0
[AUTOSAR] Lower Multiplicity	1

1.6.1.5 EthTrcv/EthTrcvConfigSet/EthTrcvConfig/EthTrcvSleepModeExecutionDelay

[RTA-BSW] Description	Specifies the time delay in seconds to execute a sleep (see OA TC10) for a Ethernet hardware (PHY), if a pending wake-up was detected while a ETH_MODE_DOWN was requested. The value shall be an integral multiple of EthTrcvMainFunctionPeriod.

[AUTOSAR] Description	Specifies the time delay in seconds to execute a sleep (see OA TC10) for a Ethernet hardware (PHY), if a pending wake-up was detected while a ETH_MODE_DOWN was requested.
[REMOVED] Introduction	The value shall be an integral multiple of EthTrcvMainFunctionPeriod.

1.6.1.6 EthTrcv/EthTrcvConfigSet/EthTrcvConfig/EthTrcvWakeUpRequestRepetitionPeriod

[RTA-BSW] Description	Specifies the repetition period in seconds of a wake-up request, if an active communication request (ETH_MODE_ACTIVE_WITH_WAKEUP_REQUEST) was triggered by the upperlayer. The value shall be an integral multiple of EthTrcvMainFunctionPeriod.
[AUTOSAR] Description	Specifies the repetition period in seconds of a wake-up request, if an active communication request (ETH_MODE_ACTIVE_WITH_WAKEUP_REQUEST) was triggered by the upperlayer.
[REMOVED] Introduction	The value shall be an integral multiple of EthTrcvMainFunctionPeriod.

1.6.1.7 EthTrcv/EthTrcvConfigSet/EthTrcvConfig/EthTrcvWakeupRequestNumberOfRepetitions

[RTA-BSW] Lower Multiplicity	0
[AUTOSAR] Lower Multiplicity	1

1.6.1.8 EthTrcv/EthTrcvConfigSet/EthTrcvConfig/EthTrcvWakeupSleepOnDatalineEnabled

[REMOVED] Introduction	If the parameter is configured, wake-up and sleep on data line is supported according to OA TC10 by the used Ethernet hardware (PHY). If the parameter is set to TRUE, the wake-up and sleep on data line functionality according to OA TC10 is enabled. If the parameter is set to FALSE, the wake-up and sleep on data line functionality according to OA TC10 is disabled. Note: Disabling of the wake-up and sleep functionality is used if Ethernet hardware (PHY) parts are connected, where one PHY do not support the OA TC10 wake-up and sleep on dataline functionality. This may needed for legacy scenarios, where ECUs are taken over from previous car lines and no OA TC10 Ethernet hardware (PHY) were used.

[RTA-BSW] Description: Specifies if wake-up on data line according to OA TC10 is supported by the used Ethernet hardware (PHY) and if the functionality (wake-up and sleep on dataline) is enabled. If the parameter is configured, wake-up and sleep on data line is supported according to OA TC10 by the used Ethernet hardware (PHY). If the parameter is set to TRUE, the wake-up and sleep on data line functionality according to OA TC10 is enabled. If the parameter is set to FALSE, the wake-up and sleep on data line functionality according to OA TC10 is disabled. Note: Disabling of the wake-up and sleep functionality is used if Ethernet hard-

ware(PHY) parts are connected, where one PHY do not support the OA TC10 wake-up and sleep on dataline functionality. This may needed for legacy scenarios, where ECUs are taken over from previous car lines and no OA TC10 Ethernet hardware (PHY) were used.

[AUTOSAR] Description: Specifies if wake-up on data line according to OA TC10 is supported by the used Ethernet hardware (PHY) and if the functionality (wake-up and sleep on dataline) is enabled.

1.6.1.9 EthTrcv/EthTrcvGeneral/EthTrcvGetBusWuReasonApi

[REMOVED] Default Value	false

1.6.1.10 EthTrcv/EthTrcvGeneral/EthTrcvMaxTrcvsSupported

[ADDED] Description	EthTrcvMaxTrcvsSupported

1.6.1.11 EthTrcv/EthTrcvGeneral/EthTrcvSetPhyTestModeApi

[RTA-BSW] Description	This api is used to configure the Phy to requested test mode. This api is available for TJA1100 and RTL9000AA EthTrcv.
[AUTOSAR] Description	Enables / Disables EthTrcv_SetPhyTestMode API.
[RTA-BSW] Lower Multiplicity	0
[AUTOSAR] Lower Multiplicity	1
[ADDED] Default Value	false

1.6.1.12 EthTrcv/EthTrcvGeneral/EthTrcvWakeUpSupport

[RTA-BSW] Description	Configures wake-up to polling or interrupt or to not used/not supported. In case no wake up is supported by the hardware, the BSWMD pre-configuration shall be set to ETHTRCV_WAKEUP_NOT_SUPPORTED.
[AUTOSAR] Description	Configures how to detect a signaled wake-up by hardware: polling, asychronous check via signal, interrupt or to not used/not supported. In case detection of a wake-up is not support (e.g. wake-up detection not needed or Ethernet hardware does not support wake-up detection), the BSWMD pre-configuration shall be set to ETHTRCV_WAKEUP_NOT_SUPPORTED.
[RTA-BSW] Enumeration Literals	ETHTRCV_WAKEUP_BY_INTERRUPT, ETHTRCV_WAKEUP_BY_POLLING, ETHTRCV_WAKEUP_NOT_SUPPORTED, ETHTRCV_WAKEUP_BY_ASYNCHRONOUS_CHECK
[AUTOSAR] Enumeration Literals	ETHTRCV_WAKEUP_BY_ASYNCHRONOUS_CHECK, ETHTRCV_WAKEUP_BY_INTERRUPT, ETHTRCV_WAKEUP_BY_POLLING, ETHTRCV_WAKEUP_NOT_SUPPORTED

1.7 API Definition

Table 1.63.EthTrcv Supported API functions

EthTrcv_Init	Initialises the module.
EthTrcv_SetTransceiverMode	Enables/disables the indexed transceiver.
EthTrcv_GetTransceiverMode	Obtains the state of the indexed transceiver.
EthTrcv_CheckWakeup	Called by EthIf if a wakeup interrupt is detected.
EthTrcv_StartAutoNegotiation	Restarts the negotiation of the transmission parameters used by the indexed transceiver.
EthTrcv_TransceiverLinkStateRequest	Request the given link state for the given transceiver
EthTrcv_GetLinkState	Obtains the link state of the indexed transceiver.
EthTrcv_GetBaudRate	Obtains the baud rate of the indexed transceiver.
EthTrcv_GetDuplexMode	Obtains the duplex mode of the indexed transceiver.
EthTrcv_SetPhyTestMode	Activates a given test mode
EthTrcv_SetPhyLoopbackMode	Activates a given loopback mode
EthTrcv_GetPhySignalQuality	Obtains the current signal quality of the link of the indexed transceiver
EthTrcv_SetPhyTxMode	Activates a given transmission mode
EthTrcv_RunCableDiagnostic	Trigger the cable diagnostics for the given Ethernet transceiver
EthTrcv_GetCableDiagnosticsResult	Retrieves the cable diagnostics result of a given transceiver
EthTrcv_GetPhyIdentifier	Obtains the PHY identifier of the Ethernet Transceiver according to IEEE
EthTrcv_GetVersionInfo	Returns the version information of this module.
EthTrcv_ReadMiiIndication	Triggered by previous Eth_ReadMii call to read out via MII interface.
EthTrcv_WriteMiiIndication	Triggered by previous Eth_WriteMii call when information has been written via MII interface.
EthTrcv_MainFunction	Used for polling state changes and wakeup reasons and calls EthIf_TrcvModeIndication when the transceiver mode changed.

For further information on these functions, please refer to AUTOSAR_SWS_EthernetTransceiverDriver.pdf

1.8 Configuration Advice

> **NOTE**
>
> Before configuring this module, please note that some of the default values used by ConfGen to produce the default ECU Configuration can be changed, and this is done by either adding a Conf-Gen Enhancer Mapping (CEM) or by adding a rule to an *algo.properties* file. For more information, see the "Configuration Generation" section of the RTA-CAR User Guide.

1.8.1 General Notes

- Only 100Mbps configuration is supported for all the Phy hardware types.
- Only pre-compile time is supported.
- Only Sync support is currently given for MDIO communication. (Async support has not currently been tested).

1.8.2 EthIfEthDriverPrefix

If the EthIfEthDriverPrefix parameter is not configured, or if it does not exist in the EthStack, then the EthTrcv_Cfg_DriverApi_cst structure will be generated by default with Eth_WriteMii and Eth_ReadMii function pointers. The definition of these APIs must be handled within the application. See the Integration Advice for more details on the EthIfEthDriverPrefix parameter.

1.8.3 Exclusive Areas

If the project implements the auto generation of exclusive areas in Schm_<Module>.h using RTE, then below configuration must be ensured:

EcuCRbRteInUse set to **True EcuCRbLegacySchMInUse** set to **False** Configure **RteBswExclusiveAreaImpl** container **RteBswEnforceExclusiveAreaApisGeneration** set to **True**

NOTE: Dependency on RTE version 12.6.0 and above.

1.9 Integration Advice

1.9.1 Init Functions

Initialization of the EthTrcv module is done by EthTrcv_Init and EthTrcv_TransceiverInit.

EthTrcv_TransceiverInit must follow EthTrcv_Int in the initialization sequence.

1.9.1.1 Error Handling during Initialization

During its initialization, EthTrcv checks for access to the Ethernet transceiver hardware because, without that access, it will not be able to initialize the physical Ethernet transceiver hardware. Some possible reasons for not having this access

include:

- Incorrect port configuration for MDIO and MDC pins.
- Incorrect initialization of the Ethernet MAC or MAC driver (Eth).
- The transceiver hardware is in an OFF state.
- The transceiver hardware is in Reset state.
- The oscillator to the transceiver hardware may not be working correctly.

If the driver does not have access, then an ETHTRCV_E_ACCESS error will be given to DEM only if DEM is configured. If DEM is not configured, then a 'ETHTRCV_E_MII_TIMEOUT_READ' error will be reported to DET if DET is configured to STD_ON.

1.9.1.2 Example Initialization

Example code for Initialization of EthTrcv:

```
void EthTrcvTest_Init (void)
{
    /* ETH L2 and L1 layer intialization */
    Eth_Init(NULL_PTR);/* Eth Driver Initialization */
    Eth_ControllerInit(0,0);/* Eth Controller Initialization */
    EthTrcv_Init(NULL_PTR);/* Eth Transceiver driver Initialization */
    EthTrcv_TransceiverInit(0,0);/* Eth Transceiver hardware Initialization */
    EthTrcv_SetTransceiverMode(0,ETH_MODE_ACTIVE);/* Transceiver mode setup */
    EthTrcv_TransceiverLinkStateRequest(0,ETHTRCV_LINK_STATE_ACTIVE);/*Transceiver Link Active
request*/
    EthTrcv_MainFunction();/* Actual mode change happens in EthTrcv_MainFunction */
    /* Wait for the mode indication from EthTrcv driver to EthIf(upper layer) */
    Eth_SetControllerMode(0, ETH_MODE_ACTIVE);/* Eth Controller mode setup */
}
```

1.9.2 DeInit Functions

No specific handling necessary.

1.9.3 Scheduled Functions

EthTrcv_MainFunction is used for polling state changes and wakeup reasons. It calls EthIf_TrcvModeIndication when the transceiver mode changes, and it stores wakeup events if EthTrcvWakeUpSupport has been set to ETHTRCV_WAKEUP_BY_POLLNG.

1.9.3.1 Eth_WriteMii and Eth_ReadMii

The Eth_ReadMii and Eth_WriteMii APIs (provided by the Eth module) must be Synchronous. The diagnostic features of EthTrcv are not supported if these Eth APIs are Asynchronous.

The Eth_WriteMii and Eth_ReadMii APIs differ between AR v4.2.2 and v4.0.3 for the Return Type, and are remapped to the functions indicated below, which maps to the relevant compatible prototype, based on your configuration.

1.9.3.2 Using the EthIfEthDriverPrefix parameter

For some specific MCAL providers, the APIs are developed with customized names. Infineon MCALs, for example, have APIs that have a prefix of <_17_Eth-

Mac_>inserted into the name (e.g. Eth_17_EthMac_ReadMii).

EthIf allows you to reference the appropriate prefix for your chosen MCAL with the EthIfEthDriverPrefix configuration parameter. If EthIfEthDriverPrefix is used, changes must be made to the integration file *EthIf_Integn.c* to reflect the prefix in API names (Extern and Definition). Some samples are shown below using an Infineon Eth Driver on a TC26x Board as an example. In this case, EthIfEthDriver-Prefix is configured as *17_EthMac*.

All EthTrcv APIs need to be remapped with syntax as: EthTrcv_6_Integ-n_(remapped to `EthTrcv_<EthIfEthDriverPrefix>`)

Note that in the current implementation of EthTrcv, these modifications have to be made manually, but in a future release this will be handled in automatic code generation.

Note also that similar modifications are required for APIs used by EthIf, see the EthIf Integration Advice for more details.

1.9.4 Update API Names (Extern)

```
/*******************************************************************************
+ Rename each API to include the Prefix. Examples:
+ Eth_6_Integn_ReadMii --> Eth_17_EthMac_ReadMii
+ Eth_6_Integn_WriteMii --> Eth_17_EthMac_WriteMii
*******************************************************************************/
Std_ReturnType Eth_17_EthMac_ReadMii( uint8 CtrlIdx,
                                      uint8 TrcvIdx,
                                      uint8 RegIdx,
                                      uint16 * RegValPtr );

Std_ReturnType Eth_17_EthMac_WriteMii( uint8 CtrlIdx,
                                       uint8 TrcvIdx,
                                       uint8 RegIdx,
                                       uint16 RegVal );
```

1.9.5 Update API Names (Definition)

```
Std_ReturnType Eth_17_EthMac_ReadMii( uint8 CtrlIdx,
                                      uint8 TrcvIdx,
                                      uint8 RegIdx,
                                      uint16 * RegValPtr )
{
        Std_ReturnType lRetVal_en = E_NOT_OK;
        /* Call adaptation : convert arguments if required and forward call to third party driver
module */
        Eth_ReadMii(CtrlIdx, TrcvIdx, RegIdx, RegValPtr);
        /* Because of mismatch of function prototype between AR 4.0.3 and AR4.2.2, this function always
return E_OK.*/
        lRetVal_en = E_OK;
        return lRetVal_en;
}
```

```
Std_ReturnType Eth_17_EthMac_WriteMii( uint8 CtrlIdx,
                                       uint8 TrcvIdx,
                                       uint8 RegIdx,
                                       uint16 RegVal )
{
        Std_ReturnType lRetVal_en = E_NOT_OK;
        /* Call adaptation : convert arguments if required and forward call to third party driver
module */
        Eth_WriteMii(CtrlIdx, TrcvIdx, RegIdx, RegVal);
        /* Because of mismatch of function prototype between AR 4.0.3 and AR 4.2.2, this function
always return E_OK.*/
        lRetVal_en = E_OK;
        return lRetVal_en;
}
```

1.9.5.1 OS Resources

The number of the active OS resources, and their usage, depends on the user configuration of the AUTOSAR System and Basic Software. It is the user's responsibility to ensure that the `ENTER` and `EXIT` resource API are implemented to disable/enable interrupts as required. Alternatively they must be configured in the OS with the correct task/ISR allocation.

The following is a list of all exclusive areas that this module might use regardless of the user configuration. It is the job of the integrator to ensure that the implementation of these exclusive areas is correctly defined. This implementation can be done using the RTE exclusive area configuration wizard or by manual implementation of the corresponding BSW API calls to ensure exclusivity.

The EthTrcv module uses the following OS resources:

- EthTrcv_MiiCommunication
- EthTrcv_RequestedMode
- EthTrcv_CurrentMode
- EthTrcv_StartAutoNegotiation
- EthTrcv_LinkState
- EthTrcv_RequestedLinkState
- EthTrcv_BaudRate
- EthTrcv_DuplexMode
- EthTrcv_ModeIndicationPendFlag
- EthTrcv_localWakeup

1.10 Hardware Requirements

Table 1.64.EthTrcv Hardware Requirements

Support	Usage
FLASH	4536 bytes
RAM	79 bytes

Usage calculated based on a sample configuration.

2 Ethernet Tranceiver 11 StdAutomPhy

Module name	EthTrcv 11 StdAutomPhy
Package	RTA-HWD (Eth)
AUTOSAR Module ID	73
AUTOSAR Version	23-11

2.1 Introduction

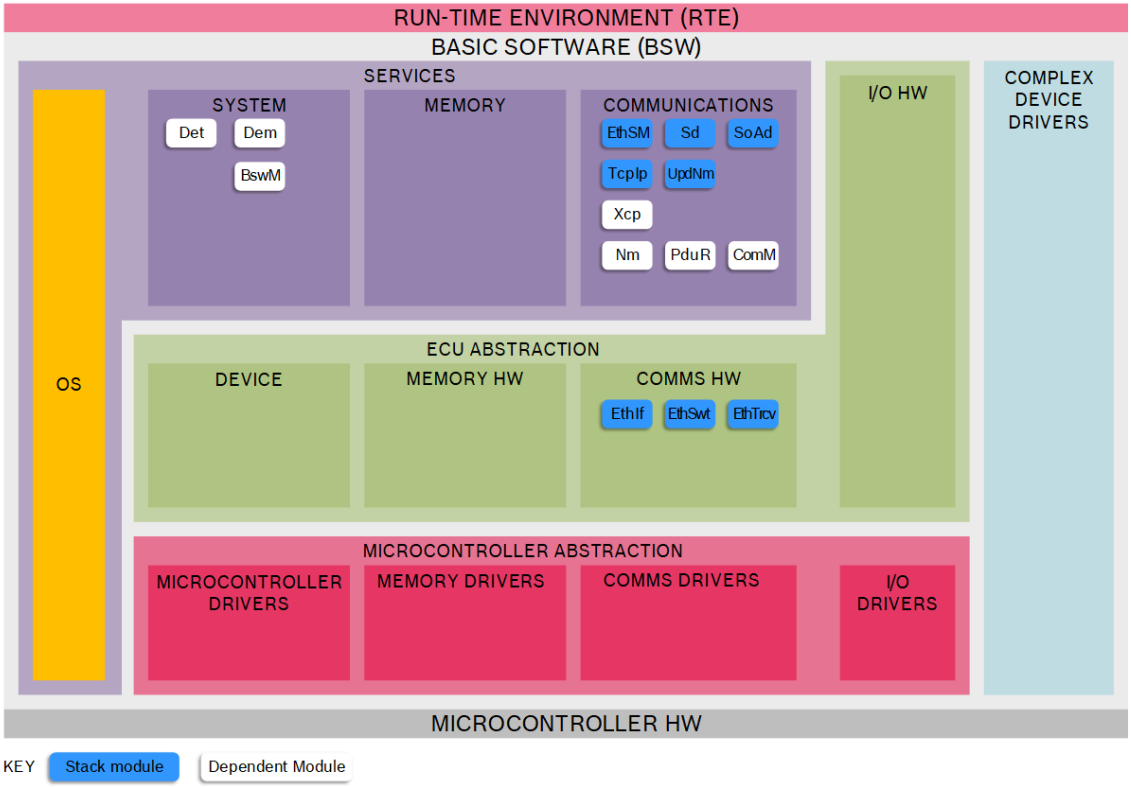

Figure 2.1.RTA-ETH Architecture

The Ethernet Transceiver Driver (EthTrcv) is located in the communication hardware abstraction layer, between the communication drivers layer and the upper communication services layer, and it forms part of the RTA-HWD (Eth) stack.

The diagram above shows the position of the EthTrcv module in the AUTOSAR layered architecture.

EthTrcv provides a hardware-independent interface comprising multiple physical Ethernet Transceivers to the upper layer Ethernet Interface (EthIf). This interface is uniform to all physical Ethernet Transceivers, allowing the upper layer to access the underlying bus system in a uniform manner.

The main tasks of the EthTrcv module include:

- Changing the state of the hardware transceiver

- Reading the current state of the hardware transceiver
- Detecting errors and the linkstate on the Ethernet line

The functionality provided by EthTrcv differs for each hardware transceiver - see the Supported Features section for more details.

2.1.1 Other Interaction/Dependencies

Eth

- The Eth driver configures the MDIO management communication path. It provides the Eth_ReadMii and Eth_WriteMii APIs for EthTrcv_11_StdAutomPhy, allowing it to access and initialize the physical Ethernet transceiver hardware.

EthIf

- The EthIf module uses EthTrcv_11_StdAutomPhy and calls its initialization APIs. In addition, EthTrcv calls the EthIf_TrcvModeIndication callback API to indicate Trcv mode changes to EthIf.

BswM

- The BSW Mode Manager (BswM) is used by the Ethernet Transceiver Driver for scheduling data consistency mechanisms and main function handling.

Port

- The Port driver configures the GPIO port pins (MDC and MDIO) used for the EthTrcv driver as input or output.

Module Configuration

This module can be generated by the RTA-BSW Configuration Generator.

For any unsupported automatic configuration, the module can be manually configured by the user.

2.2 Supported Features

- Getting/Setting the mode of the transceiver
- Checking the wakeup mode of the transceiver
- Monitoring the link status
- Auto Negotiation
- Getting the Baud Rate
- MDIO interactions (incl. Async)
- Running the diagnostic services

> **NOTE**
>
> Some functionality of EthTrcv is hardware-dependent and may vary according to the vendor-specific transceiver type being accessed.

2.2.1 Supported Hardware Devices

EthTrcv_11_StdAutomPhy supports the following hardware devices:

Manufacturer	Device Name
Marvell	88Q2220M version B0
Marvell	88Q2220M version B2

See the limitations section for information on the diagnostic features that are not currently supported by the various devices.

2.3 Limitations

2.3.1 Hardware-Specific Limitations

- MARVELL 88Q2220M (Tested with B0 and B2 samples):
  - Multi-Intances: Not tested.
  - DEM error reporting or calling DEM error API is not supported.
- If EthIfEthDriverPrefix parameter is configured then Eth_6_SynopsisQos_WriteMiii and Eth_6_SynopsisQos_ReadMii will be generated

2.4 Exclusions

The following features from AR v22-11 are not supported in version v12.7.0.

API functions not currently implemented:

- `EthTrcv_GetMacMethod`
- `EthTrcv_MacSecUpdateSecY`
- `EthTrcv_MacSecInitRxSc`
- `EthTrcv_MacSecResetRxSc`
- `EthTrcv_MacSecAddTxSa`
- `EthTrcv_MacSecUpdateTxSa`
- `EthTrcv_MacSecDeleteTxSa`
- `EthTrcv_MacSecAddRxSa`
- `EthTrcv_MacSecUpdateRxSa`
- `EthTrcv_MacSecDeleteRxSa`
- `EthTrcv_MacSecGetTxSaNextPn`
- `EthTrcv_MacSecGetMacSecStats`

- `EthTrcv_MacSecSetControlledPortEnabled`
- `EthTrcv_EthSwitchMacSecUpdateSecY`
- `EthTrcv_EthSwitchMacSecAddTxSa`
- `EthTrcv_SwitchMacSecAddRxSa`
- `EthTrcv_SwitchMacSecGetMacSecStats`
- `EthTrcv_SetTransceiverWakeupMode`
- `EthTrcv_GetTransceiverWakeupMode`
- `EthTrcv_SetPhyLoopbackMode`
- `EthTrcv_SetPhyTxMode`
- `EthTrcv_GetPhyIdentifier`

The following configuration parameters from AR v22-11 are not supported in version v12.7.0.

- There are no known exclusions in this version.

2.5 API Definition

For further information on these functions, please refer to AUTOSAR_SWS_EthernetTransceiverDriver.pdf

2.6 Configuration Advice

NOTE

Before configuring this module, please note that some of the default values used by ConfGen to produce the default ECU Configuration can be changed, and this is done by either adding a Conf-Gen Enhancer Mapping (CEM) or by adding a rule to an *algo.properties* file. For more information, see the "Configuration Generation" section of the RTA-CAR User Guide.

2.6.1 Required Configuration Changes

- The EthTrcvRbHardwareType should be set to STDAUTOMPHY_MARVELL_88Q2220M or STDAUTOMPHY_MARVELL_88Q2221M.
- EthTrcvRbSplitInitIntoBlocks should be set to FALSE if the initialization is done in the init runnable and there is no runtime constrains. (init will be done in one block in context of EthIf_MainFunctionState→EthTrcv_11_StdAutomPhy_TransceiverInit)
- When the init is in one block this can take more than 10ms, therefore EthTrcvRbSplitInitIntoBlocks should be set to TRUE if executing EthIf_MainFunctionState→EthTrcv_11_StdAutomPhy_TransceiverInit from a 10ms task.

2.6.2 Configuration Interface

The BSW SWS allows several configuration classes:

- Pre-compile time - Currently only this is supported.
- Link time
- Post-build time

The standardization of configuration parameters does not necessarily fix the configuration class of parameters. Instead it may define different configuration class implementation variants of a module. However, only pre-compile time configuration is supported by the Ethernet Transceiver driver module. Postbuild and link build time are not supported by the EthTrcv driver.

2.7 Integration Advice

2.7.1 Init Functions

Initialization of the EthTrcv_11_StdAutomPhy module is done by EthTrcv_11_StdAutomPhy_Init and EthTrcv_11_StdAutomPhy_TransceiverInit.

EthTrcv_11_StdAutomPhy_TransceiverInit must follow EthTrcv_11_StdAutomPhy_Init in the initialization sequence.

2.7.1.1 Error Handling during Initialization

During its initialization, EthTrcv checks for access to the Ethernet transceiver hardware because, without that access, it will not be able to initialize the physical Ethernet transceiver hardware. Some possible reasons for not having this access include:

- Incorrect port configuration for MDIO and MDC pins.
- Incorrect initialization of the Ethernet MAC or MAC driver (Eth).
- The transceiver hardware is in an OFF state.
- The transceiver hardware is in Reset state.
- The oscillator to the transceiver hardware may not be working correctly.

2.7.1.2 Example Initialization

Example code for Initialization of EthTrcv_11_StdAutomPhy:

```
{
    /* ETH L2 and L1 layer intialization */
    Eth_Init(NULL_PTR);/* Eth Driver Initialization */
    Eth_ControllerInit(0,0);/* Eth Controller Initialization */
    EthTrcv_11_StdAutomPhy_Init(NULL_PTR);/* Eth Transceiver driver Initialization */
    EthTrcv_11_StdAutomPhy__TransceiverInit(0,0);/* Eth Transceiver hardware Initialization */
    EthTrcv_11_StdAutomPhy_SetTransceiverMode(0,ETH_MODE_ACTIVE);/* Transceiver mode setup */
    EthTrcv_11_StdAutomPhy_TransceiverLinkStateRequest(0,ETHTRCV_LINK_STATE_ACTIVE);/*Transceiver Link
Active request*/
    EthTrcv_11_StdAutomPhy_MainFunction();/* Actual mode change happens in EthTrcv_MainFunction */
    /* Wait for the mode indication from EthTrcv driver to EthIf(upper layer) */
    Eth_SetControllerMode(0, ETH_MODE_ACTIVE);/* Eth Controller mode setup */
}
```

2.7.2 DeInit Functions

No specific handling necessary.

2.7.3 Scheduled Functions

EthTrcv_MainFunction is used for polling state changes and wakeup reasons. It calls EthIf_TrcvModeIndication when the transceiver mode changes, and it stores wakeup events if EthTrcvWakeUpSupport has been set to ETHTRCV_WAKEUP_BY_POLLNG.

2.7.3.1 Eth_WriteMii and Eth_ReadMii

The Eth_ReadMii and Eth_WriteMii APIs (provided by the Eth module) must be Synchronous. The diagnostic features of EthTrcv are not supported if these Eth APIs are Asynchronous.

The Eth_WriteMii and Eth_ReadMii APIs differ between AR v4.2.2 and v4.0.3 for the Return Type, and are remapped to the functions indicated below, which maps to the relevant compatible prototype, based on your configuration.

2.7.3.2 Using the EthIfEthDriverPrefix parameter

For some specific MCAL providers, the APIs are developed with customized names. Infineon MCALs, for example, have APIs that have a prefix of <_17_EthMac_> inserted into the name (e.g. Eth_17_EthMac_ReadMii).

EthIf allows you to reference the appropriate prefix for your chosen MCAL with the EthIfEthDriverPrefix configuration parameter. If EthIfEthDriverPrefix is used, changes must be made to the integration file *EthIf_Integn.c* to reflect the prefix in API names (Extern and Definition). Some samples are shown below using an Infineon Eth Driver on a TC26x Board as an example. In this case, EthIfEthDriverPrefix is configured as *17_EthMac*.

All EthTrcv APIs need to be remapped with syntax as: EthTrcv_6_Integn_(remapped to `EthTrcv_<EthIfEthDriverPrefix>`)

Note that in the current implementation of EthTrcv, these modifications have to be made manually, but in a future release this will be handled in automatic code generation.

Note also that similar modifications are required for APIs used by EthIf, see the EthIf Integration Advice for more details.

2.7.3.3 OS Resources

The number of the active OS resources, and their usage, depends on the user configuration of the AUTOSAR System and Basic Software. It is the user's responsibility to ensure that the `ENTER` and `EXIT` resource API are implemented to disable/enable interrupts as required. Alternatively they must be configured in the OS with the correct task/ISR allocation.

The following is a list of all exclusive areas that this module might use regardless of the user configuration. It is the job of the integrator to ensure that the imple-

mentationof these exclusive areas is correctly defined. This implementation can be done using the RTE exclusive area configuration wizard or by manual implementation of the corresponding BSW API calls to ensure exclusivity.

The EthTrcv_11_StdAutomPhy module uses the following OS resources:

- EthTrcv_11_StdAutomPhy_MiiCommunication
- EthTrcv_11_StdAutomPhy_RequestedMode
- EthTrcv_11_StdAutomPhy_CurrentMode
- EthTrcv_11_StdAutomPhy_LinkState
- EthTrcv_11_StdAutomPhy_RequestedLinkState
- EthTrcv_11_StdAutomPhy_ModeIndicationPendFlag
- EthTrcv_11_StdAutomPhy_Wakeup
- EthTrcv_11_StdAutomPhy_NumActiveWakeUp
- EthTrcv_11_StdAutomPhy_ActiveWakeupTriggered
- EthTrcv_11_StdAutomPhy_WakeupRepitionTimeout

2.8 Hardware Requirements

Information relating to the hardware resources required by this module is not currently available.

3 Glossary

DUT
Device Under Test

Eth Ethernet Driver

EthIf
AUTOSAR Ethernet Interface module

EthTrcv
Ethernet Transceiver Driver

GPIO
General Purpose Input/Output

MAC
Media Access Control

MCAL
Microcontroller Abstraction Layer

MDC
Management Data Clock

MDIO
Management Data Input/Output

MII Media Independent Interface

OUI
Organizationally Unique Identifier

PTP
Prevision Time Protocol

RMII
Reduced Media Independent Interface

SQI Signal Quality Information